

CUSTOMER SEGMENTATION + RFM ANALYSIS

Project Domain:	E-Commerce / Retail Analytics
Dataset:	500 Customers 7,000 Transactions FY 2022–2023
Tools Used:	Excel Power Query Power Pivot DAX RFM Model
Analysis Date:	31-December-2023 (Max Transaction Date)
Segments Found:	9 RFM Segments identified
Level:	Big 4 Consulting Standard

1. WHAT IS RFM ANALYSIS?

RFM stands for **Recency, Frequency, and Monetary** — a proven marketing framework used by top consulting firms (McKinsey, BCG, Deloitte, PwC) to segment customers based on their purchasing behaviour. It answers the most important business question: *which customers are most valuable, and what should we do with each segment?*

Metric	Definition	Business Meaning	Scoring Logic
RECENCY (R)	Days since last purchase (from Analysis Date)	Recent buyers are more likely to buy again	Lower days = Higher score ≤30 days = Score 5
FREQUENCY (F)	Total number of orders placed by customer	Frequent buyers show strong engagement	More orders = Higher score ≥20 orders = Score 5
MONETARY (M)	Total revenue generated by customer (FinalRevenue)	High spenders drive disproportionate revenue	Higher spend = Higher score ≥2L = Score 5

2. RFM SEGMENT DEFINITIONS

Each customer is scored 1–5 on R, F, M. The combination of scores determines their segment. This project identified **9 active segments** from the dataset:

Segment	RFM Score	Behaviour	Business Action
Champions	R≥4, F≥4, M≥4	Buy often, spent most, bought recently	Reward + Advocacy program
Loyal Customers	R≥3, F≥4, M≥3	Regular buyers, consistent spenders	Upsell + Cross-sell
Potential Loyalists	R≥4, F≥2, M≥3	Recent + multiple purchases	Nurture to Loyalty
New Customers	R≥5, F≤2	Bought recently, low frequency	Onboarding + 2nd purchase push
Promising	R≥3, F≥3, M≥3	Recent, below avg spend	Activation + education
Need Attention	R≤2, F≥3, M≥3	Above avg R/F/M but inactive	Re-engage urgently
About to Sleep	R≥3, F≤2, M≥2	Becoming inactive — danger zone	15% discount + FOMO campaign
At Risk	R≤2, F≥2, M≥2	Used to buy — now quiet	Win-back email + survey
Lost	R≤2, F≤2	Long inactive, low value	Minimal investment

3. KEY FINDINGS FROM DATASET

Segment	Customers	Revenue (₹)	Rev %	Avg Recency	Avg Freq
Champions	200	3,75,97,812	49.5%	17.6 days	17.3
Loyal Customers	207	3,03,57,687	40.0%	28.6 days	12.7
Potential Loyalists	40	33,91,594	4.5%	22.7 days	11.4
About to Sleep	25	20,70,409	2.7%	55.8 days	7.8
New Customers	19	15,98,359	2.1%	13.7 days	8.3
Need Attention	4	4,95,924	0.7%	132.8 days	12.5
Promising	3	2,37,749	0.3%	94.7 days	10.7
At Risk	1	1,25,867	0.2%	127.0 days	9.0
Lost	1	46,354	0.1%	5.0 days	10.0
TOTAL	500	7,59,21,756	100%	25.9 days	14.0

4. BUSINESS QUESTIONS & ANSWERS

These are the exact questions a Big 4 interviewer or business stakeholder will ask. Learn these answers — they form the core of your project presentation.

Q1. What is the Pareto principle in this dataset? Is 80/20 proven?

Answer: Yes — fully proven. Champions (40% of customers) contribute 49.5% of revenue. Top 2 segments (Champions + Loyal Customers) together = 89.5% of total revenue from just 81.4% of customers. This validates the classic Pareto principle and justifies focused retention investment on these two segments.

Q2. Which customers are most valuable and what should we do with them?

Answer: Champions (200 customers, ₹3.76Cr revenue) are the most valuable. They buy frequently, recently, and spend the most. Action: Launch a VIP Loyalty Program, offer early access to new products, referral bonuses, and exclusive events. Goal is to retain them AND convert them into brand advocates.

Q3. Which segment poses the highest immediate revenue risk?

Answer: 'About to Sleep' segment — 25 customers with ₹20.7L revenue — have stopped buying (avg 55.8 days since last purchase) but were previously active. This is the most actionable at-risk group. Recommended action: Personalised re-engagement email with 15% discount + FOMO messaging within 30 days.

Q4. What does Recency tell us about customer health?

Answer: Champions have avg recency of just 17.6 days — they bought very recently. In contrast, Need Attention segment has avg 132.8 days and At Risk has 127 days. This 7–8x recency gap proves that time-since-purchase is the strongest predictor of churn. Any customer beyond 90 days with no purchase needs immediate intervention.

Q5. How does Frequency differentiate customer segments?

Answer: Champions average 17.3 orders compared to Lost customers' 10 orders — a 1.7x gap. However, the real gap is in RECENCY — Lost customers actually have low recency (5 days) suggesting they may be new lost customers. The key frequency insight: Loyal Customers at 12.7 orders vs About to Sleep at 7.8 orders shows that frequency is declining before customers fully disengage.

Q6. Which product category should get the highest marketing investment?

Answer: Electronics (₹1.75Cr, 23.1%) and Sports (₹1.70Cr, 22.4%) are the top two categories together contributing 45.5% of revenue. For Champion and Loyal segment re-engagement campaigns, these categories should be the primary offer focus. Beauty (₹1.45Cr) and Clothing (₹1.44Cr) are close behind — suitable for cross-sell campaigns.

Q7. Which geography should be prioritised for marketing spend?

Answer: Germany leads with ■1.46Cr (19.2%) followed by India (■1.36Cr, 17.9%) and Canada (■1.33Cr, 17.5%). These 3 markets contribute 54.6% of total revenue. Marketing budget allocation recommendation: Germany 25%, India 20%, Canada 20%, UK 15%, USA 15%, France 5%.

Q8. What retention strategy should be implemented for each segment?

Answer: Champions: VIP program + referral incentives. Loyal: Upsell to premium. Potential Loyalists: Membership trial + personalised emails. New Customers: Onboarding series + 2nd purchase offer. About to Sleep: 15% discount win-back. Need Attention: Personalised re-engagement + survey. At Risk: High-touch outreach + large discount. Lost: Annual email only — minimal investment.

Q9. Why is FinalRevenue used instead of Revenue for Monetary calculation?

Answer: FinalRevenue = Revenue × (1 - Discount), which represents the actual money received after discounts. Using gross Revenue would overstate the value of discount-heavy customers. In a Big 4 context, Monetary should always reflect net realised value, not gross billing. This ensures RFM scores accurately reflect true customer profitability contribution.

Q10. What is the business impact of this RFM analysis?

Answer: This analysis enables data-driven customer strategy: (1) Protect ■7.93Cr Champions + Loyal revenue through targeted retention. (2) Recover ■21.96L at-risk revenue through win-back campaigns. (3) Convert 40 Potential Loyalists into Loyal Customers — potential ■3.4Cr additional revenue. (4) Optimise marketing spend by focusing 90% budget on top 3 segments.

5. EXCEL SKILLS DEMONSTRATED IN THIS PROJECT

Skill	Where Used	Why Important
Power Query	Data import, cleaning, merging tables	Automated ETL — no manual data prep
Star Schema	4-table data model with relationships	Industry-standard data modelling
DAX Measures	KPIs: Revenue, Customers, Avg Frequency	Dynamic calculations in Power Pivot
MAXIFS	Last purchase date per customer	Multi-condition aggregation
COUNTIF/SUMIF	Segment Summary table	Dynamic aggregation by category
AVERAGEIF	Avg Recency + Frequency by segment	Conditional averaging
IFS Formula	R, F, M scoring (1-5)	Multi-condition logic
IFERROR	Protect AVERAGEIF from #DIV/0!	Error handling in formulas
Conditional Format	Segment Summary — Color scales	Visual data storytelling
Pivot Tables	All 6 dashboard charts as data source	Dynamic summarisation
Pivot Charts	6 visualisations on Dashboard	Professional data presentation
UNIQUE()	Dynamic segment list in summary	Spill formula — auto-updates

6. HOW TO REPLICATE THIS PROJECT

Step 1 — Data Import

Import Transactions, Customers, Products tables via Power Query. Clean data types, handle nulls, enrich Transactions with Delivery Category, Profit column. Load all to Data Model (connection only).

Step 2 — Star Schema

In Power Pivot Diagram View, create relationships: Transactions[CustomerID] → Customers[CustomerID], Transactions[ProductID] → Products[ProductID], Transactions[InvoiceDate] → Calendar[Date].

Step 3 — RFM Calculations

Create RFM_Analysis sheet. Use Power Pivot to get Max InvoiceDate, Distinct Count, Sum of FinalRevenue per CustomerID. Add Recency = MAX_DATE - Last_Purchase. Score each metric 1–5 using IFS formula.

Step 4 — Segmentation

Add Segment column using IFS + AND formula combining R, F, M scores. 9 segments: Champions to Lost.

Step 5 — Segment Summary

Create Segment_Summary sheet using COUNTIF, SUMIF, AVERAGEIF + IFERROR. Add Revenue % and Cumulative % columns for Pareto analysis.

Step 6 — Power Pivot Integration

Merge RFM Segments back into Customers table via Power Query Merge. Now Segment available in DAX.

Step 7 — DAX Measures

Create measures: Total Revenue, Total Customers, Total Orders, Avg Order Value, Champions Revenue %, At Risk Count.

Step 8 — Dashboard

Create 6 Pivot Tables, add 6 Pivot Charts (Horizontal Bar, Donut, Grouped Column). Build Dashboard sheet with KPI cards, insights, and 3 slicers connected to all pivots.